

Ahmad Raza

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SUMMARY

Software Engineering undergraduate and Deep Learning practitioner building production multi-agent RAG systems with local LLM inference. Research focus on Cognitive-Load-Aware Educational Assistants and hardware-efficient agentic architectures. Combines rigorous theoretical foundations in NLP and Statistical Learning Theory with hands-on expertise in Generative AI.

EDUCATION

Thal University

Bachelor of Science in Software Engineering

Bhakkar, Punjab

Oct. 2022 – May 2026

- **CGPA:** 3.83/4.0
- Relevant Coursework: Software Engineering, Data Structures & Algorithms, Web Engineering, Database Systems, Cloud Computing

PROJECTS

Sage (FYP) | *LangChain, LangGraph, llama.cpp, FastAPI, React* | [GitHub](#)

Feb. 2026 – Present

- Architected an offline-first Multi-Agent Academic Assistant featuring a **7-agent LangGraph** graph (Router, Retrieval, Reasoning, Quiz, Diagram, Planner, Research, Code-Fix).
- Engineered a hardware-aware **llama.cpp** lifecycle manager with dynamic context-window scaling (3k–64k) and VRAM-based GPU offloading; applied **BayesOpt** (Optuna) for CPU compiler flags, cutting server cold-start from **15–18 s to 3–5 s** (PP ~116 tok/s, TG ~12.5 tok/s).
- Built a modular **tool suite** (sandboxed code execution, symbolic calculator, Mermaid diagram renderer & validator, web/academic search, and library-docs retriever) integrated as LangChain tools across agents.
- Developing a **desktop application** (pywebview, FastAPI) featuring real-time **SSE streaming** and a persistent **Long+Short Term memory** targeting fully offline, zero-cloud deployment.

Seq Modeling From Scratch | *Python, NumPy, Math* | [GitHub](#)

Nov. 2025

- Implemented **RNN, LSTM, GRU** without auto-differentiation frameworks to master gating mechanisms and computational graphs.
- Manually derived and coded forward and backward passes (Backpropagation Through Time) to manage gradient flow in dynamic networks and enable training.
- Built **Seq2Seq** models with **Bahdanau** and **Luong attention** mechanisms specifically for language modeling tasks. Trained character and word-level **language models** to demonstrate practical sequence learning.

NeuroScope | *Python, NumPy, Matplotlib* | [GitHub](#) | [PyPI](#) | [Docs](#)

Sep 2025

- Engineered a production-grade Python **framework** for informed training and diagnosis of neural networks.
- Developed a real-time diagnostics system monitoring **10** health indicators (e.g., Dead Neurons, Gradient SNR, Weight Update Ratios) and a comprehensive analytics and evaluation system validated against research by Glorot, He, and Pascanu.
- Designed dynamic visualizations for neuron activation, gradients, and weights histograms and evolution plots to track training dynamics.
- Maintained **60%+** test coverage and automated **PyPI** publishing workflows via GitHub Actions.

TECHNICAL WRITING

A Synthesis of Biological and Computational Learning | [Read Article](#)

Dec. 2025

- Authored a comprehensive analysis connecting synaptic plasticity (LTP/STDP) with Backpropagation and Transformers.
- Synthesized literature from Hebb (1949) to Vaswani (2017) to explore the Information Bottleneck Theory.
- Examined the divergence between biological efficiency ($\approx 20W$) and current LLM training paradigms.

TECHNICAL SKILLS

Deep Learning & ML: LLMs, Transformers, Fine-tuning, RAG, Multi-Agent Systems, Sequence Models, Representation Learning, Statistical Learning Theory

Foundations: Linear Algebra, Multivariate Calculus, Probability Theory, Statistics, Optimization

MLOps & Infra: FastAPI, MLflow, DVC, Model Deployment & Serving, Docker, Kubernetes, CI/CD, AWS

Languages & Libraries: Python, SQL, NumPy, Matplotlib, Seaborn, Scikit-Learn, XGBoost, Optuna, PyTorch, LangChain, LangGraph

CERTIFICATIONS

DeepLearning.AI (Coursera): Deep Learning Specialization, Machine Learning Specialization, Mathematics for Machine Learning & Data Science Specialization

Other Specializations: MLOps Specialization (Duke University), IBM Machine Learning Professional Certificate